

Eden Gilbert Kiseka

Embedded Systems Developer & Technical Educator

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📍 Kampala, Uganda

EXPERIENCE

Backend Engineer

Googah Goats Limited

Oct 2025 – Present
Kampala, Uganda

- Architect backend infrastructure using Google AppScript and JavaScript.
- Design automation systems and data workflows managing business operations.
- Build custom APIs and integration scripts connecting various services.
- Develop understanding of distributed systems and API design.

Technical Instructor – Embedded Systems

Neriko Electronics

Oct 2024 – Sep 2025
Kampala, Uganda

- Taught embedded systems programming to students using C, C++, and Arduino platform.
- Developed comprehensive curriculum covering microcontroller fundamentals through advanced IoT applications.
- Designed hands-on laboratory exercises demonstrating real-world embedded applications.
- Covered topics: GPIO programming, UART/SPI/I2C communication, ADC/DAC interfacing, PWM control, interrupt handling, timer programming.
- Instructed students in industry tools: PlatformIO (embedded development), TinkerCAD (circuit simulation), Fritzing (PCB design), Proteus (circuit simulation), AutoCAD Electrical (professional schematics).
- Created educational materials: lecture notes, circuit diagrams, code examples, project specifications.
- Mentored students individually through debugging hardware-software integration challenges.
- Taught systematic troubleshooting using multimeters, oscilloscopes, and logic analyzers.

PROJECTS

kore – Schrödinger-Poisson Solver

Rust, PyO3, C ABI, NumPy, RustFFT, Rayon

[View Repository](#)

- Portable pseudo-spectral kernel for fuzzy dark matter simulation workloads.
- Implements split-step Fourier evolution with RustFFT and Rayon threading for parallelism.
- Exposes a stable C ABI and `pykore` Python bindings via PyO3/NumPy, bridging Python workflows with HPC-grade performance.
- Verified with Rust integration tests, strict C11 smoke builds, and Python tests covering in-place complex128 stepping with GIL release on the compute path.

Amber – Android Reconstruction Pipeline

Kotlin, DEX IR, SSA, JavaPoet, Gradle

[View Repository](#)

- Lifts APK and DEX artifacts into typed IR, CFG, SSA, and generated source, then synthesizes a complete Android Studio project around the result.
- Combines dex parsing, Java/Kotlin emission, Gradle synthesis, metadata analysis, and evidence-rich diagnostics to turn reverse engineering into a structured engineering workflow.
- Supports end-to-end reconstruction: emitted Java baselines, Gradle project generation, metadata-aware Kotlin recovery, Compose detection, and resource pipeline integration.
- Open source under Apache 2.0; CI-driven validation on real APKs.

TECHNICAL SKILLS

Embedded Programming	C, C++, Arduino	Software Development	Python, Kotlin, Java,
Microcontroller Tools	Microchip Studio, PlatformIO, Arduino IDE		JavaScript, Rust, Swift
Communication Protocols	UART, SPI, I2C, GPIO, PWM	Digital Design	Logisim, FPGA (learning), RISC-V, Logic Simulation
Peripherals	ADC/DAC, Timers, Interrupts, Sensors	Hardware Description	Verilog/SystemVerilog (learning)
Circuit Design	TinkerCAD, Fritzing, Proteus, AutoCAD Electrical	Version Control	Git, GitHub, GitLab
Debug Tools	Multimeter, Oscilloscope, Logic Analyzer	Development IDEs	VS Code, PlatformIO, Microchip Studio, Arduino IDE
Platforms	Arduino (Uno, Mega, Nano), ESP32/ESP8266, STM32	Operating Systems	Linux, Windows
Embedded Linux	Raspberry Pi, Cross-Compilation, Device Drivers	Build Systems	Makefiles, CMake, PlatformIO

EDUCATION

Associate Degree in Electrical Engineering & Computer Science	2025 – 2027
University of the People	United States
Diploma in Electronics and Electrical Engineering	2025 – 2027
Uganda Institute of ICT	Uganda

LANGUAGES

English	Fluent	Luganda	Fluent	French	Intermediate (B2)
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